

# RF HiRes Dash — Technical Reference

Propagation Physics, Formulas & Calculation Methods

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## 1. ITU-R P.368-9 Propagation Model

ITU-R Recommendation P.368-9 is an international technical standard for predicting ground wave propagation for frequencies from 0.15 MHz to 30 MHz (LF, MF, and HF bands).

### What is ITU-R P.368-9?

"Ground wave" means the radio signal travels close to the Earth's surface, following its curvature. This dominates at short-to-medium distances (<1,000 km) and low-to-medium frequencies, making it ideal for broadcasting, maritime, and amateur radio.

### Valid Frequency Range

0.15 MHz to 30 MHz

- 0.15–0.535 MHz: LF (low frequency)
- 0.535–1.7 MHz: MF (medium frequency, AM broadcast)
- 1.7–30 MHz: HF (high frequency, amateur, maritime, aviation)

For frequencies outside this range, use alternative models (e.g., UHF/VHF line-of-sight models for 30+ MHz).

### Key Inputs to P.368-9

- Frequency (MHz): 0.15–30
- Transmitter height (m): antenna height above ground
- Receiver height (m): antenna height above ground
- Radiated power (W): effective radiated power
- Ground conductivity (S/m): electrical conductivity at TX and RX
- Distance (km): TX to RX distance
- Elevation profile: terrain heights along the path

### Output

Received field strength: E (dBμV/m) — decibels above 1 microvolt per meter. This is the standard metric for RF coverage.

## 2. Path Loss Fundamentals

### Free Space Path Loss

In an ideal environment with no obstacles, path loss follows:

$$L_{fs}(dB) = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.45$$

where: d = distance in km, f = frequency in MHz

*Example: 10 MHz, 100 km in free space*

$$L_{fs} = 20 \log(100) + 20 \log(10) + 32.45 = 40 + 20 + 32.45 = 92.45 \text{ dB}$$

### Practical Path Loss (Ground Wave)

Real-world path loss includes attenuation due to ground interaction:

$$L_{path}(dB) = L_{fs} + A(d, f, \sigma, \epsilon_r, h_{tx}, h_{rx})$$

where: A = additional attenuation from ground losses, diffraction, and antenna effects (typically 5–50 dB depending on terrain and conductivity). P.368-9 computes A numerically using complex wave number and Sommerfeld integrals — it is an iterative calculation, not a simple formula.

## 3. Medium Frequency (MF/LF) Propagation (0.15–30 MHz)

## Frequency Bands

- 0.15–0.535 MHz: LF (Low Frequency)
- 0.535–1.6 MHz: MF (Medium Frequency, AM broadcast)
- 1.6–30 MHz: HF (High Frequency, amateur, maritime, aviation)

## The Ground Wave Mechanism

At frequencies below 30 MHz, radio signals propagate predominantly as ground waves — energy that travels close to the Earth's surface, following its curvature. This occurs because of:

- Refraction in the lower atmosphere (bending the signal downward)
- Electrical interaction with the Earth's surface (which supports wave propagation)
- Signal energy that can reach beyond line-of-sight

The ground wave dominates for short-to-medium distances (<1,000 km) and remains constant day and night, unlike skywave (ionospheric bounce), which varies with atmospheric conditions.

## Ground Conductivity ( $\sigma$ ) — The Critical Parameter

Electrical conductivity measures how easily current flows through a material. Higher  $\sigma$  = better propagation (lower path loss).

| Medium       | Conductivity (S/m) |
|--------------|--------------------|
| Sea water    | 5.0                |
| Wet soil     | 0.01 – 0.1         |
| Average land | 0.005              |
| Dry sand     | 0.0001             |

Ground conductivity is the MOST IMPORTANT parameter affecting path loss (other than distance and frequency). A 10× change in  $\sigma$  can change range by ±30%.

## Why Separate Conductivity at TX and RX?

- TX over water ( $\sigma = 5$  S/m), RX over land ( $\sigma = 0.005$  S/m): dramatically different propagation
- TX in urban area (higher moisture), RX rural: conductivity varies
- Mixed paths: P.368-9 averages conductivity along the path using Millington integration

## ITU-R P.368-9 Millington Calculation Method

The P.368-9 model is NOT a simple formula. Instead, it:

- Uses Sommerfeld integral theory to model wave propagation over lossy ground
- Integrates the complex wave number accounting for ground conductivity
- Applies Millington mixed-path integration for paths with varying ground types
- Includes antenna correction factors (height and frequency dependent)
- Computes diffraction over terrain obstacles from the elevation profile

## RF HiRes Dash Implementation

- Input: Frequency, TX/RX heights, antenna ERP, ground conductivity ( $\sigma_{tx}$ ,  $\sigma_{rx}$ ), elevation profile (~200 points from SRTM)
- Calculation: Numerical integration of P.368-9 equations
- Output: Field strength E (dB $\mu$ V/m) at each RX location
- Computational complexity:  $O(N)$  where  $N \approx 200$  terrain profile points

## Terrain Diffraction in the MF Band

Knife-edge diffraction loss is calculated from the elevation profile using Fresnel zone clearance at the dominant obstacle. At MF frequencies, diffraction is less pronounced than at VHF/UHF but still significant for large terrain obstacles (hills >100 m relief).

## 4. VHF/UHF Propagation (30 MHz – 2 GHz)

### Frequency Range & Applications

- 30–300 MHz: VHF (Very High Frequency, FM broadcast, mobile)
- 300 MHz–3 GHz: UHF (Ultra High Frequency, TV, cellular, WiFi)

RF HiRes Dash is validated to 2 GHz; use above 2 GHz experimentally only.

### Why Ground Wave Fails at VHF/UHF

- Wavelength becomes very short (10 m at 30 MHz; 15 cm at 2 GHz)
- Ground conductivity effects negligible relative to free-space loss
- Direct and reflected rays dominate over guided ground wave

As a result, P.368-9 is NOT used above 30 MHz. Instead, the model uses Fresnel-Kirchhoff diffraction + two-ray theory.

### Component 1: Fresnel-Kirchhoff Diffraction (Deygout Method)

This models how terrain obstacles block radio waves.

- Identify the dominant obstacle from the elevation profile
- Calculate Fresnel parameter:  $v = \sqrt{2} \times (2d_1d_2)/(\lambda D)$  where  $d_1, d_2$  = obstacle distances to TX/RX,  $D$  = total distance,  $\lambda$  = wavelength
- Apply Fresnel-Kirchhoff loss:  $\text{Loss (dB)} = \max(0, 10 \log(0.5 + 0.62v))$
- For multiple obstacles, apply successive diffraction (Bullington method)

Result: Diffraction loss typically 5–50 dB. Higher frequencies → shorter wavelengths → larger diffraction loss. Doubling frequency increases loss by ~6 dB.

### Component 2: Two-Ray Ground Reflection

Models constructive and destructive interference between the direct ray (straight path) and the reflected ray (bounces off ground). Phase difference creates constructive interference (enhancement) or destructive interference (fading nulls).

### Component 3: Radio Horizon & Horizon Diffraction

Radio horizon distance (approximate):  $d_{\text{horizon}} \text{ (km)} \approx 4.12 \times \sqrt{(h_{\text{tx}} \text{ [m]} + h_{\text{rx}} \text{ [m]})}$

Beyond-horizon paths: signal must diffract over Earth's bulge, producing 20–60 dB loss depending on frequency and height difference.

### Component 4: VHF Thickness Penalty

Thickness penalty (dB) =  $0.4 \times (f \text{ [MHz]} / 30) \times \sqrt{(\text{diffraction loss [dB]})}$

Effect: Frequency doubling (e.g., 100 MHz → 200 MHz) increases loss by ~3 dB.

### Combined Path Loss

- LOS paths: Path Loss = Free Space Loss + Two-Ray Factor + Diffraction (if partial)
- Non-LOS paths: Path Loss = Free Space Loss + Deygout Diffraction + Horizon Loss + VHF Thickness Penalty

## 5. Terrain Elevation Data & Processing

SRTM (Shuttle Radar Topography Mission)

- Resolution: 90 m (3 arc-seconds)
- Vertical accuracy: ±16 m (90% confidence)
- Coverage: ±60° latitude (most continents, not polar regions)
- Source: NASA/USGS public domain

Elevation Profile Extraction & Processing

- Step 1: Define great-circle path from TX to RX
- Step 2: Fetch SRTM elevation along the path (typically 4 tiles per continental path)
- Step 3: Resample using 200-point contour-enhanced algorithm — preserves ridges, valleys, and inflection points while reducing computational cost
- Step 4: Pass 200-point profile to propagation engine (P.368-9 for MF/LF; Deygout for VHF/UHF)

Elevation Data Fallback

- 1. Try OpenTopography API (slower, remote call)
- 2. If OpenTopography also unavailable: Use flat terrain (h = sea level)

■ Flat terrain assumption degrades accuracy significantly (±5–10 dB error possible in hilly regions). Always check SRTM coverage before planning in remote areas.

6. Frequency & Validation Range

| Range        | Band      | Status             | Applications                    |
|--------------|-----------|--------------------|---------------------------------|
| 0.01–0.3 MHz | VLF/LF    | Valid; sparse SRTM | LW broadcast, LF nav beacons    |
| 0.3–30 MHz   | LF/MF/HF  | Fully validated    | AM broadcast, maritime, amateur |
| 30–2,000 MHz | VHF/UHF   | Fully validated    | FM, mobile, TV, cellular        |
| 2–30 GHz     | Microwave | Experimental only  | 5G, Ku/Ka satellite, mmWave     |

The 30 MHz Boundary

- < 30 MHz: Use P.368-9 Millington (ground wave + diffraction)
- ≥ 30 MHz: Use Fresnel-Kirchhoff (LOS + diffraction + two-ray)

This boundary reflects the physical transition from ground-wave-dominated to LOS-dominated propagation and aligns with ITU-R P.368-9's specified range.

## 7. Antenna Polarisation — Types, Gain & System Loss

RF HiRes Dash uses effective radiated power (ERP) as its primary input. ERP already embeds antenna gain relative to the reference — the app does not apply antenna gain internally. Polarisation effects on path loss are handled via the System Loss parameter.

### 7.1 Antenna Types Supported

| Antenna Type             | Band     | Model Applied                   | Notes                                |
|--------------------------|----------|---------------------------------|--------------------------------------|
| Vertical Monopole        | LF/MF/HF | ITU-R P.368-9 ground wave       | Primary model. Vertically polarised. |
| Horizontal Dipole (NVIS) | HF only  | Near Vertical Incidence Skywave | Not recommended above 30 MHz.        |
| Vertical Monopole        | VHF/UHF  | Fresnel-Kirchhoff + two-ray     | Default for VHF/UHF.                 |

### 7.2 Antenna Gain Reference

The following gains apply at low elevation angles relevant to coverage planning. ERP must be entered relative to the same reference the app uses for the selected band.

| Antenna                                            | Gain (dBi) | vs $\frac{1}{2}\lambda$ Dipole (dBd) | Notes                                         |
|----------------------------------------------------|------------|--------------------------------------|-----------------------------------------------|
| Isotropic (reference)                              | 0.00       | −2.15                                | Theoretical reference only                    |
| $\frac{1}{2}\lambda$ Vertical Dipole               | 2.15       | 0.00                                 | Standard broadcast reference                  |
| $\frac{1}{4}\lambda$ Vertical Monopole (ideal gnd) | 5.19       | +3.04                                | Requires perfect infinite ground plane        |
| $\frac{1}{4}\lambda$ Vertical Monopole (real gnd)  | 2–5        | −0.15 to +2.85                       | Practical; ground plane finite                |
| $\frac{1}{2}\lambda$ Horizontal Dipole             | 2.15       | 0.00                                 | Same gain as vertical dipole; pattern differs |
| Circular Polarisation (CP)                         | 2.15       | 0.00                                 | Combination of two orthogonal linear elements |

### 7.3 Circular Polarisation (CP)

CP is not directly modelled. At LF/MF ground wave frequencies, the horizontal component attenuates almost completely within a few wavelengths — CP effectively becomes vertically polarised at any useful distance.

**Recommended approach:** Select Vertical Monopole. Set ERP to the vertical component only. For a true CP antenna with equal power split, subtract 3 dB via System Loss.

### 7.4 Horizontal Polarisation

Horizontal ground wave propagation is not supported. At LF/MF frequencies, the horizontal component attenuates far more rapidly than the vertical component over ground — horizontal polarisation is not viable for ground wave coverage planning.

At VHF/UHF, horizontal polarisation is supported in the sense that ERP is accepted for any antenna orientation, but polarisation mismatch loss at the receiver is not modelled internally — it must be accounted for via System Loss (see Section 7.6).

■ Horizontal Dipole (NVIS) mode models Near Vertical Incidence Skywave for HF only. It is not recommended above 30 MHz and does not model horizontal ground wave.

### 7.5 System Loss — What It Covers

System Loss (dB) is subtracted from ERP before any propagation calculation. It is a catch-all for all passive losses between the transmitter output and the radiated field. Use 0 if ERP is already measured at the antenna terminals after all losses.

**System Loss should include:**

- Feedline and cable losses between transmitter output and antenna
- Connector and switch losses
- Antenna earth mat / ground system efficiency losses (particularly relevant for LF/MF monopoles where ground system quality significantly affects radiation efficiency)
- Polarisation mismatch loss at the receiver (see Section 7.6)

## 7.6 Polarisation Mismatch Loss — Add to System Loss

Polarisation mismatch loss depends on the transmission path, distance, and environment. Multipath and terrain scattering progressively depolarise the signal, reducing mismatch loss at longer paths. Urban environments depolarise faster than open rural terrain.

### Ground Wave (LF/MF)

| Scenario                  | System Loss to Add                                |
|---------------------------|---------------------------------------------------|
| V TX → V RX               | 0 dB                                              |
| H TX → any RX             | Not viable — H ground wave non-propagating        |
| Any TX → any RX (skywave) | 0–3 dB (Faraday rotation randomises polarisation) |

### VHF/UHF — Direct/LOS Path

| Scenario                       | System Loss to Add |
|--------------------------------|--------------------|
| V TX → V RX                    | 0 dB               |
| V TX → H RX, <5 km open        | 15–20 dB           |
| V TX → H RX, 5–50 km           | 8–12 dB            |
| V TX → H RX, >50 km / urban    | 3–6 dB             |
| V TX → H RX, beyond horizon    | 1–3 dB             |
| CP TX → Linear RX, LOS         | 3 dB               |
| CP TX → Linear RX, multipath   | 1–3 dB             |
| CP TX → CP RX (same sense)     | 0 dB               |
| CP TX → CP RX (opposite sense) | >>20 dB            |

■ These are planning estimates. Real-world polarisation mismatch loss varies with antenna pattern, ground reflection angle, and local scattering environment. For precise link budgets, measure with the actual antenna system.

## 7.7 Skywave Accuracy Warning

The skywave figure shown in PtP results is a simplified first-order estimate — a single-hop E-layer model using 100 km reflection height and 0.3 reflection coefficient. It is intended as an order-of-magnitude indicator only.

### Not accounted for:

- Time of day — ionospheric conditions vary greatly; MF skywave at night can be 30–40 dB stronger than daytime
- Season and solar activity / sunspot cycle
- Geomagnetic conditions and D-layer absorption
- Multi-hop paths; MUF/LUF
- Faraday rotation (polarisation randomised by ionosphere — mismatch loss 0–3 dB regardless of antenna type)

### For regulatory submissions or interference analysis, use:

- ITU-R P.1147 — MF skywave prediction

- ITU-R P.533 / VOACAP — HF skywave prediction